

MATH1014 Algebra

London Wallace

Semester 2, 2026

Contents

Preface	v
0.1 What is Algebra?	v
0.2 Table of Symbols	v
1 Fields	1
1.1 Lecture 1	1
1.1.1 Binary Operations	1
1.1.2 Axioms	1
1.1.3 The Definition of a Field	1
1.1.4 $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Q}$ are Fields	2
1.1.5 $\mathbb{Z}$ is not a Field	2
1.1.6 The Binary Field	2
1.1.7 The Properties of Fields	3
1.1.8 Power Sets	4
1.2 Lecture 2	4

Preface

THESE NOTES follow, and are meant to cement the course *Algebra*, taught at the University of Western Australia by Dr. Gordon Royle in Semester 2 of 2026. Where needed, the content of lectures is supplemented by *Linear Algebra Done Right* by Sheldon Axler.

0.1 What is Algebra?

I DO NOT YET KNOW ENOUGH TO SAY.

0.2 Table of Symbols

Symbol	Meaning
$\forall$	For every <i>or</i> for all
$\exists$	There exists
$\in$	In <i>or</i> is an element of
$\subset$	Is a proper subset of
$\subseteq$	Is a subest of
$\cup$	Union with
$\cap$	Intersection of
$\setminus$	Not <i>or</i> without
$\emptyset$	Empty set
$>$	Greater than
$<$	Less than
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$ n $	The absolute value of n
$\square$	It has been proven

Chapter 1

Fields

1.1 Lecture 1

1.1.1 Binary Operations

BINARY OPERATIONS are *operations*¹ that take two inputs and produce one output. Addition (+) and multiplication ($\times$) are common examples of binary operations.

A unary operation is one that takes only one input, and produces one output; for example, sine. The function $f(x, y, z) = x + y + z$ is an example of a *ternary* operation. There are an infinite amount of n -ary operations, however binary operations form the basis of the study of fields and rings.

1.1.2 Axioms

AXIOMS are statements, assumed to be true, used to define the domain of a particular mathematical theory. The goal is to find the smallest set of obviously true statements that define the domain which we are interested in. If any axioms are logical consequences of others, they should be omitted, and likewise if any properties of the underlying structure we wish to define are lost, an axiom should be added to its definition to provide those properties. Axioms form a basis for further reasoning. We cannot prove anything without first agreeing on the postulates of our theories.

1.1.3 The Definition of a Field

A FIELD is a *set* S , equipped with the two binary operations $+$ and $\times$, that behave similarly to how addition and multiplication would behave on real numbers, that also satisfies the following *axioms*:

¹Operations are not yet defined.

1. Commutativity of $+$. $[a + b = b + a]$
2. Associativity of $+$. $[a + (b + c) = (a + b) + c]$
3. Existence of the additive identity 0. $[a + 0 = a]$
4. Existence of the additive inverse. $[\forall a \in S, \exists b \in S : a + b = 0]$
5. Commutativity of $\times$. $[a \times b = b \times a]$
6. Associativity of $\times$. $[a \times (b \times c) = (a \times b) \times c]$
7. Existence of the multiplicative identity 1. $[a \times 1 = a]$
8. Existence of the multiplicative inverse. $[\forall a \in S, \exists b \in S : a \times b = 1]$
9. Distributivity. $[a \times (b + c) = a \times b + a \times c]$
10. Non-triviality. $[0 \neq 1]$

1.1.4 $\mathbb{R}, \mathbb{C}$, and $\mathbb{Q}$ are Fields

STARTING WITH THE REALS, it is quite clear that all the axioms are met. For any number $a \in \mathbb{R}$, its additive inverse would be $-a$, and its multiplicative inverse would be a^{-1} . The identities represented by the symbols 0 and 1 above are in fact 0 and 1, and axioms 1, 2, 5, 6, 9, and 10 are obvious. The same is true for the rationals and the complex numbers.

1.1.5 $\mathbb{Z}$ is not a Field

NOT EVERY SUBSET of a field is a field. The integers are not a field because it is not true that for any $a \in \mathbb{Z}$, $\exists b \in \mathbb{Z}$ with $a \times b = 1$. 2 is certainly an integer, but for $2 \times b = 1$ to hold, $b = \frac{1}{2}$, which is certainly not an integer!

1.1.6 The Binary Field

THE SET $\{0, 1\}$ with multiplication and addition as usual, except that $1 + 1 := 0$ is a field. It is the smallest field and the unique field with two elements. Addition defined in this way is known in computer science as XOR.

1.1.7 The Properties of Fields

FROM THE AXIOMS there is a standard set of proofs used to establish the properties of fields usually taken for granted.

1. Uniqueness of 0.

The additive identity 0 is unique. Let 0 and $0'$ be additive identities, then $0 + 0' = 0$ and $0' + 0 = 0'$. From the commutativity of addition, $0 + 0' = 0' + 0$, and thus $0 = 0'$. $\square$

2. Uniqueness of 1.

Let 1 and $1'$ be multiplicative identities, then $1 \times 1' = 1$ and $1' \times 1 = 1'$. But $1 \times 1' = 1' \times 1$, thus $1 = 1'$. $\square$

3. Uniqueness of the inverses.

Let b and c be additive inverses of a , then

$$b = b + 0$$

$$b = b + (a + c)$$

$$b = (a + b) + c$$

$$b = 0 + c$$

$$b = c$$

$\square$

Let b and c be multiplicative inverses of a , then

$$b = b \times 1$$

$$b = b \times (a \times c)$$

$$b = (a \times b) \times c$$

$$b = 1 \times c$$

$$b = c$$

$\square$

4. Cancellation laws

If $a + b = a + c$, then $b = c$, and if $a \neq 0$ and $ab = ac$ then $b = c$.

5. Arithmetic of 0.

$$a \times 0 = 0$$

6. Sign rules

$$(-1) \cdot a = -a \text{ and } (-a) \cdot b = -(ab) = a \cdot (-b)$$

7. No zero divisors

If $ab = 0$ then $a = 0$ or $b = 0$. A zero divisor is a non zero element a , such that $\exists b \neq 0$ with $ab = 0$.

1.1.8 Power Sets

A POWER SET $P(S)$ is the set of all subsets of S . The set $S = \{1, 2, 3\}$ has $P(S) = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}, \emptyset\}$. If we define multiplication to be $A \cap B$ and addition to be $A \cup B \setminus A \cap B$, then all the axioms of a field are met *except* the existence of a multiplicative inverse. The multiplicative identity is the set $\{1, 2, 3\}$ since the intersection of any member of $P(S)$ with $\{1, 2, 3\}$ is itself, but it is not possible for every member of $P(S)$ to intersect with $\{1, 2, 3\}$ and produce $\{1, 2, 3\}$. A structure that satisfies every axiom of a field except the existence of a multiplicative inverse is called a *commutative unital ring*.

1.2 Lecture 2